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Multimodal representation learning over heterogeneous networks for tag-based music retrieval

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ABSTRACT

Learning how to represent data represented by features obtained from multiple modalities through representation learning strategies has received much attention in Music Information Retrieval. Among several sources of information, musical data can be represented mainly by features extracted from acoustic content, lyrics, and metadata that concentrate complementary information and have relevance when discriminating the recordings. In this work, we propose a new method for learning multimodal representations structured as a heterogeneous network capable of incorporating different musical features in constructing a representation and exploring the similarity simultaneously. Our multimodal representation is centered on the information of tags extracted from a state-of-the-art neural language model and, in a complementary way, the audio represented by the melspectrogram. We submitted our method to a robust evaluation process composed of 10,000 queries with different scenarios and model parameter variations. Besides, we compute the Mean Average Precision and compare the representation proposed to representations built only with audio or tags obtained from a pre-trained neural model. The proposed method achieves the best results in all evaluated scenarios and emphasizes the discriminative power of multimodality can add to musical representations.

1. Introduction

Determining the similarity between music pieces is an important challenge for different Music Information Retrieval (MIR) tasks. A good criterion of musical similarity is also crucial for many commercial platforms to improve users' engagement and experiences (Cheng et al., 2020), such as recommender systems (Wang et al., 2018), music query by examples (Watanabe & Goto, 2019), and automatic playlist generation (Álvarez et al., 2020). Usually, the literature uses acoustic features to represent the songs and to calculate musical similarity. Methods that exploit audio features to compute similarities have well-known limitations for music retrieval since human subjects consider several other music characteristics to interpret the similarity, like emotional, social, and cultural factors — which are hardly represented by audio features (Korzeniowski et al., 2020; Paul & Kundu, 2020; Robinson et al., 2020). To overcome these limitations, social tags have been used successfully to enrich musical data representation showing that tag-based music retrieval have potential to be more accurate than audio-based methods in some tasks (Ibrahim, Epure et al., 2020; Ibrahim, Royo-Letelier et al., 2020; Park et al., 2018).

More recently, deep learning-based models have been actively explored for music auto-tagging tasks (Ibrahim, Royo-Letelier et al., 2020; Lee et al., 2020; Lin & Chen, 2020; Lin et al., 2018; Pons et al., 2018). The main idea is to train deep neural networks on a large music dataset to automatically determine new music tags, reducing the dependence on human-generated tags. For example, Musicnn (Pons & Serra, 2019) is a model trained from 200k audio signals that can be used as an out-of-the-box music tagger. Although auto-tagging is a promising alternative, this approach is limited to a fixed tag vocabulary (Law et al., 2010). The auto-tagging process is very dependent on the training dataset, where only the most popular tags are used because they contain enough data for training deep models (Kleć & Wieczorkowska, 2021). In real-world applications, we can use out-of-vocabulary tags, as the new tags or tags do not know in the training step (Won et al., 2021). It is desired that the music retrieval task be flexible and allow queries with a virtually unlimited number of tags.

We present a multimodal representation learning from heterogeneous networks, which are beneficial for modeling data that has multiple features to build a representation to address the aforementioned limitations. This type of network may represent different objects as vertices, while network links represent the relationships between these objects. Our proposed heterogeneous network contains two types of

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objects: audio and tags. Network links indicate whether a tag is associated with a particular music. Using the melspectrogram as feature to represent the audio-content, our heterogeneous network also has links to connect the most similar pieces of music. Thus, objects representing untagged music can be reached by tag objects through walks in the heterogeneous network. We explore the heterogeneous network's topology to learn a new vector-space representation for music, i.e., an embedding space in which we can calculate similarity. This representation is centered on neural language models from the existing tag data. Our method generates a tag-based embedding space considering the links between objects in the heterogeneous network as constraints during representation learning. In short, our main contributions are threefold:

- In the context of tag-based music retrieval, we mitigate the problem of musics without associated tags using similarity between musics in a multimodal vector space via heterogeneous networks, generated by both audio features and textual tag features.
- Our multimodal representation for music data is centered on tag features extracted from neural language models. In particular, we use the state-of-the-art BERT-based language model to generate initial tag embeddings in our representation learning method. Embeddings are further refined considering the heterogeneous network topology as constraints.
- Our method enables embedding generation for any query by leveraging the neural language model. Thus, there is no out-of-vocabulary limitation as occurs in recent auto-tagging methods.

Our proposed method's performance in the *Million Song Subset* (Bertin-Mahieux et al., 2011) and *Jamendo Dataset* (Bogdanov et al., 2019) benchmarks data set. A statistical analysis considering different levels of missing tags to simulate the long-tail scenario reveals that our proposal outperforms approaches that use audio features and auto-tagging to deal with music without tags. Furthermore, we present some examples to illustrate how our method handles out-of-vocabulary tags.

2. Related work

This section discusses previous solutions to problems caused by the absence of tags in musical data sets. In addition, we explore approaches about the representation of tags in semantic embedding space, tag propagation, and the use of heterogeneous networks to perform music retrieval based on tags.

We can imagine many tags that a piece of music can have, considering the unique way that we observe its content. However, the most popular tags attributed to the track are associated with some of these categories: genre, location, feeling, and instrument (Bertin-Mahieux et al., 2010). Even considering the great variety of tags these categories have, there is a high-frequency tendency to use a small set of tags for a large music set (Wang et al., 2018). The applications, which attempt to solve MIR tasks using data sets that have an unbalanced distribution of target tags, explore strategies that aim to fill the lack of information by searching for tags from several source (Lin et al., 2016; Turnbull et al., 2009), and also by complementary information obtained from the audio content (Pons et al., 2018) as strategies to assign tags automatically (Ibrahim, Royo-Letelier et al., 2020).

There are indications that the tag category influences the choice of feature to represent the music data, where textual and audio features reach different results depending on the tag. In tasks such as emotion detection (An et al., 2017; Rachman et al., 2020), the lyrics are more related to the tags than the audio information. However, features extracted from audio have been more used in genre classification problems (Akella & Moh, 2019; Choi et al., 2018; Gupta et al., 2021). Using feature information from different sources to learn a musical representation helps supply some information when exploring the semantic complementarity between the features. Besides, this is relevant to build multimodal representations that incorporate more semantic information of the music to increase its discrimination capacity (Guo et al., 2019; Li et al., 2019; Zhang et al., 2020).

In our proposal, we consider the complementarity between audio and tags. The audio information is represented by a set of physical or perceptual features extracted directly from its content (Alías et al., 2016). Knowing the type of descriptive information that each feature has, we need to select the one related to the problem's context and use it to represent the songs (Ng et al., 2020; Pandeya & Lee, 2020). On the other hand, tag information consists of texts that can compose a semantic representation (Choi et al., 2019). However, they need to go through a processing step, usually using natural language processing techniques (Simonetta et al., 2019). The recent neural language models like Word2Vec (Çoban & Karabey, 2017; Won et al., 2021) and GloVe (Abdillah et al., 2020; Akella & Moh, 2019) are often used for this task. Due to the great diversity of musical tags' categories and vocabulary, these models can build a consistent representation (Sandouk & Chen, 2016).

In addition to exploring the strategies for representing the tags, we need to look for alternatives to fix missing tags. Tag propagation is a technique that can be applied to create additional tags in different data types to fill missing tags (Sordo et al., 2007). This technique makes it possible to balance tags' distribution in a data set when finding music with similar content and sharing their tags (Zhu & Ghahramani, 2002). The biggest challenge in this task is defining how to measure the similarity between the songs through adopted features that have a semantic relationship with the tags and perform tag information sharing coherently. To execute the propagating tags, we can measure the similarity from any information present in other music, such as acoustic features (Yang et al., 2012), presence in related playlists (Lin et al., 2018), or information set that contains contextual data, artist, listener, and other metadata (Lin & Chen, 2020).

Musical data, with multimodal structure, have modalities composed of features with specific semantic information that contribute to the perception and identification of the data (Wu et al., 2021). Instead of analyzing the features manually to build a representation for multimodal data, we can think of learning methods capable of producing latent embedding space that concentrate the semantic relations between the modalities and condense them as a unique representation (Baltrušaitis et al., 2018; Bengio et al., 2013; Rahate et al., 2022). In our problem scenario, we want to learn a representation from the audio and tags, label the music with no tags, and retrieve a similar track from a query. Thus, a possible model structure capable of encompassing this scenario's needs is the structure of heterogeneous networks. Due to the ability to incorporate different features of the data during the construction of its topology, learning models formed by heterogeneous networks can explore tag propagation and information retrieval problems when they have links between the close or similar nodes, enabling access to them.

Exploring the proximity between songs using audio information and tags to build a better measure of similarity was also an option of Nanopoulos and Karydis (2011) in its Know Thy Neighbor (KTN) method, which aims to solve a cold-start problem in MIR. The work of Oramas et al. (2017) divided the learning process to create models trained with specific features. After the models' fusion, they showed that using combined acoustic and textual information presents a superior performance of using them separately in the artist's and song recommendation problems. Solutions that look for strategies to balance the number of tags already existing as (Craw et al., 2015; Levy & Sandler, 2009; Silva et al., 2014) present themselves as a more simple functional option concerning the learning process, which can be taken as a reference for approaches related to the propagation of tags.

Although the problem structure can be associated with a network structure, this type's approaches still need further exploration in the literature. In general, the works mentioned above have chosen to use models based on neural network architectures or similar neighborhoods. However, they have a greater need for data volume for training

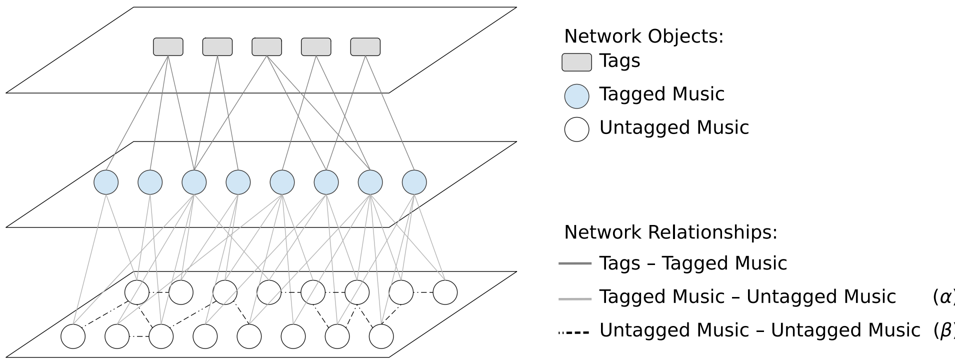

Fig. 1. Relationships between different types of objects. The heterogeneous network regularization aims to fill in missing modalities in objects. Objects on the central layer have relationships with objects on other layers. Untagged objects have relationships among objects in the same layer, constructed from the audio content. When completing the network regularization, all objects receive information from neighbors with different modalities from the network topology. The α represents the relationship number among one tagged object and other untagged objects. This parameter variation impact the propagation of tagged information. The β represents the number of the relationships among untagged objects. This parameter impacts the number of untagged objects that will receive tag information.

or higher computational cost or need complementary strategies to meet different problem settings that we want to explore simultaneously, such as cold-start and long-tail scenarios. Furthermore, the process we adopt to represent the tags allows us to reproduce problems regardless of the context of the labels.

3. Music Representation Learning over Heterogeneous Networks (MRLHN)

In this section, we present the proposed MRLHN method that explores both modalities of audio and text features to improve tag-based music retrieval tasks. In Section 3.1, we first describe the technique for generating the heterogeneous network that models relationships between songs and tags, followed by Section 3.2 that presents the process of music representation learning from the heterogeneous network.

3.1. Heterogeneous network for text and audio modalities

Let $N = (O, R, W)$ be a heterogeneous network where O represents a set of objects, R represents the relations $r_{o_i,o_j} \in R$ between objects $o_i \in O$ and $o_j \in O$ and W represents the relation weights. Our proposed heterogeneous network has three types of objects $O = \{O_L \cup O_U \cup O_T\}$: tagged music O_L, untagged music O_U, and tags O_T. The main step in data modeling via heterogeneous networks is the strategy for generating relationships between objects. In the proposed method we use three types of relationships:

- **Relationships between Tagged Music and Tags:** we directly generate $r_{o_l,o_t} \in R$ relationships between tagged music $o_l \in O_L$ and their respective tags $o_t \in O_T$. This relationship represents the textual modality of the music dataset.
- **Relationships between Tagged Music and Untagged Music:** for each untagged music $o_u \in O_U$, we calculate a total of α most similar tagged music objects $O_l \in O_L$ according to an audio-based distance measure, thereby generating a relationship $r_{o_l,o_u} \in R$ based on the audio modality. This relationship provides relevant paths in the heterogeneous network between tags and untagged music, which are explored during the representation learning process.
- **Relationships between Untagged Music:** the relationship $r_{o_u,o_u} \in R$ is obtained by calculating a total of β most similar untagged music $o_u \in O_u$ for the other untagged music through the audio modality. The goal of this relationship is to increase the number of music objects to be reached by some network path between tag objects and untagged music objects.

Fig. 1 shows an illustrative example of the proposed heterogeneous network, highlighting the three different types of objects and relationships. Note that the network structure is directly impacted by the α and β parameters, which define the degree of the music objects. Another important factor is the choice of the audio feature to calculate the relationships between songs through the audio modality, as well as the respective distance measure. In our method, the weight of the relationships is the same for all types, i.e., we use a binary weight matrix W to indicate whether the relationship between two objects exists or not. In addition, we argue that the α and β parameters are experimental, depending mainly on the diversity of the songs in the dataset (see the Experimental Evaluation section).

3.2. Tag-based music representation learning

We aim to map all different object types and relationships of the heterogeneous network into a single vector-space model. Different previous studies report that music representation learning from multi-modalities usually outperforms each representation of a single modality in musical data (Guo et al., 2019; Li et al., 2019; Pons et al., 2018; Zhang et al., 2020). Moreover, we note that some modalities may be more associated with some musical categories and present greater discriminative power in different tasks (Akella & Moh, 2019; An et al., 2017; Choi et al., 2018; Rachman et al., 2020). In the case of tag-based music retrieval tasks, we investigated whether there is a more significant relevance between text or audio modalities and whether incorporating both results in a multimodal representation with greater discriminative power than single modalities. Thus, our MRLHN method leverages the pre-trained BERT-based neural language model to obtain an initial vector space considering the existing tags. BERT-based language models are currently state-of-the-art for natural language processing tasks. BERT generates contextual word embeddings, in which the embedding of a word depends on the neighboring words, which is useful for tags composed of multiple words. Another advantage is that BERT is pre-trained on a large textual corpus, which can be seen as a transfer learning strategy when BERT is fine-tuned for other tasks.

The problem of music representation learning can be defined as the learning a mapping function $f : o_i \rightarrow \mathbf{z}_{o_i} \in \mathbb{R}^d$, where $\mathbf{z}_{o_i}$ is the learned vector of the object $o_i \in O$ in the network $N(O, R, W)$, and d is the dimension of the learned music representation space. The f mapping should preserve the heterogeneous network relationships, i.e., neighboring objects in the heterogeneous network (via direct relationships or shared neighborhoods) must have high similarity in the learned music space, while objects that do not have paths between them in the heterogeneous network must have low similarity.

We investigated a graph regularization framework as a basis for MRLHN representation learning. First, we extract BERT-based embeddings $\mathbf{b}_{o_t} \in \mathbb{R}^d$ for each tag object $o_t \in O_T$, according to some pre-trained language model. Then, these embeddings are used as an initial representation, which will be adjusted considering the objects and relationships of the heterogeneous network. Two assumptions are the basis of our learning process: (1) neighboring objects must have similar embeddings; (2) tag-type objects must preserve their initial embeddings.

Eq. (1) defines the objective function to be minimized by the MRLHN to learn the new $\mathbf{Z} \in \mathbb{R}^d$ space, in which all heterogeneous network objects are mapped.

$$\begin{aligned} Q(\mathbf{Z}) = & \frac{1}{2} \sum_{o_u \in O_U} \sum_{o_l \in O_L} w_{o_u,o_l} (\mathbf{z}_{o_u} - \mathbf{z}_{o_l})^2 \\ & + \frac{1}{2} \sum_{o_i,o_j \in O_U} w_{o_i,o_j} (\mathbf{z}_{o_i} - \mathbf{z}_{o_j})^2 \\ & + \frac{1}{2} \sum_{o_l \in O_L} \sum_{o_t \in O_T} w_{o_l,o_t} (\mathbf{z}_{o_l} - \mathbf{z}_{o_t})^2 \\ & + \lim_{\mu \to \infty} \mu \sum_{o_t \in O_T} (\mathbf{z}_{o_t} - \mathbf{b}_{o_t})^2 \end{aligned} \tag{1}$$

We aim to learn a $\mathbf{Z}$ space where the distances $(\mathbf{z}_{o_u} - \mathbf{z}_{o_l})^2$ between untagged music $o_u \in O_U$ and tagged music $o_l \in O_L$ are small if they are linked in the heterogeneous network, $w_{o_u,o_l} > 0$, as represented in the first term $\frac{1}{2} \sum_{o_u \in O_U} \sum_{o_l \in O_L} w_{o_u,o_l} (\mathbf{z}_{o_u} - \mathbf{z}_{o_l})^2$. Similarly, we also want to minimize the distances between untagged musics, according to the second term $\frac{1}{2} \sum_{o_i,o_j \in O_U} w_{o_i,o_j} (\mathbf{z}_{o_i} - \mathbf{z}_{o_j})^2$. The third term $\frac{1}{2} \sum_{o_l \in O_L} \sum_{o_t \in O_T} w_{o_l,o_t} (\mathbf{z}_{o_l} - \mathbf{z}_{o_t})^2$ aims to obtain embeddings of tagged musics $o_l \in O_L$ close to the embeddings of their related tags $o_t \in O_T$. While the first three terms of the objective function are related to the first assumption, the last term aims to guarantee the second assumption, thereby minimizing the distance between the tag embeddings from $\mathbf{Z}$ space and the initial tag embeddings from BERT language model. The μ tending to infinity in practice penalizes the objective function if these distances are large.

Eq. (1) can be seen as a particular case of graph regularization through embedding propagation, which has well-known theoretical proofs of convergence and minimization through convex optimization methods. In this paper, we explore an iterative and scalable solution based on label propagation to handle large graphs, which obtain solutions close to the convex optimization methods.

Algorithm 1: *Embedding Propagation for Music Representation Learning*

Input: Heterogeneous Network $N = (O, R, \mathbf{W})$, Initial Embeddings $\mathbf{B}$, Parameter $\hat{\mu}$
Output: Music Embeddings $\mathbf{Z}$

```
1 begin
2     Compute the degree matrix D ∈ ℝ^{|O|×|O|}, where D_ii = Σ_{j=1}^{|O|} W_ij
3     Compute the transition probability matrix P = D^{-1/2} W D^{-1/2}
4     repeat
5         Z(t + 1) = μ̂ P Z(t) + (1 − μ̂) B
6     until convergence;
7 end
8 return Z
```

Algorithm 1 shows the MRLHN pseudocode. MRLHN receives as input the heterogeneous network, the initial tag embeddings, and the μ parameter. In line 2, we calculate the matrix $\mathbf{D}$, which indicates the degree of each object in the network. This matrix is used to calculate a transition probability matrix between the network objects (line 3), considering the possible paths and edge weights. In Lines 4–6, the $\mathbf{Z}$ space is learned considering the initial $\mathbf{B}$ embeddings. In each iteration, the embeddings are smoothly propagated to all objects in the network. The μ parameter indicates how much we preserve the tags' initial embeddings. Convergence is obtained when there are no significant changes between embeddings after some iterations, or when a maximum number of iterations is reached.

4. Experimental results

In our experiments, we used the *MillionSongSubset*[1] and *Jamendo Dataset* (Bogdanov et al., 2019) as benchmark dataset. The *MillionSong-Subset* is a subset obtained from the original Million Song Dataset that has 10,000 pieces of music organized by social tags and other various metadata related to pieces of music and artists. The number of unique tags is 726 in different categories, and they are present individually or combined with other tags. We consider only tagged music to explore the tag-based retrieval problem. Thus, we removed the untagged samples, leaving 3708 pieces of music. The Jamendo Dataset is an open dataset for music tagging with 55,701 songs tagged, at least one of 623 tags in several categories like genre, instrument, and mood/theme.

4.1. Experimental setup

We defined the following scenario for evaluating the proposed method. Given the task of obtaining the top-k songs according to an input tag, we simulate the absence of tags returning only the top-m songs most similar to the query, with $m < n$. The rest of the $n - m$ songs are obtained considering three approaches:

- **Audio-based similarity:** represents approaches usually used to generate playlists, information retrieval, and recommendation systems. We use the well-known melspectrogram generated from the music's audio signal as a feature, extracted using the library Librosa[2] with default parameters.
- **Auto-tagging:** represents recent approaches that explore deep learning to infer music tags. We used the pre-trained Musicnn model (Pons & Serra, 2019) and selected the top-10 tags with the highest confidence score. Musicnn has a vocabulary of 50 tags, composed of popular tags also existent in the Million Song Dataset.
- **MRLHN:** short for Music Representation Learning over Heterogeneous Networks represents the method proposed in this paper. We evaluated the generation of the heterogeneous network considering the following parameters: $\alpha = \{3, 5, 7\}$ and $\beta = \{3, 5, 7\}$. For the audio modality, we use the melspectrogram of the first 30 s of the audio signal represents by your spectral centroid. For the textual modality, we used DistilBERT Multilingual to generate the existing tags' initial embeddings, and the representation learning step was carried out by minimizing Eq. (1). The source code of the proposed method is available at https://github.com/AngeloMendes/MRLHN for the experiments' reproducibility.

We experimented with 200 main queries composed of social tags available in each dataset benchmark for evaluation. These tags were chosen because they occur on at least ten recordings. We use a reference model where all songs have associated tags generating the ground truth and calculating the Mean Average Precision MAP@k, where k is the number of pieces to be returned. MAP@k is close to 1 when the retrieved music list is similar to the ground truth list (reference model); otherwise, it is close to 0. The MAP@k metric is formulated in Eq. (2).

$$MAP = \frac{1}{|K|} \sum_{k=1}^{K} AP_k \tag{2}$$

[1] http://millionsongdataset.com/pages/getting-dataset/.
[2] https://librosa.org/doc/.

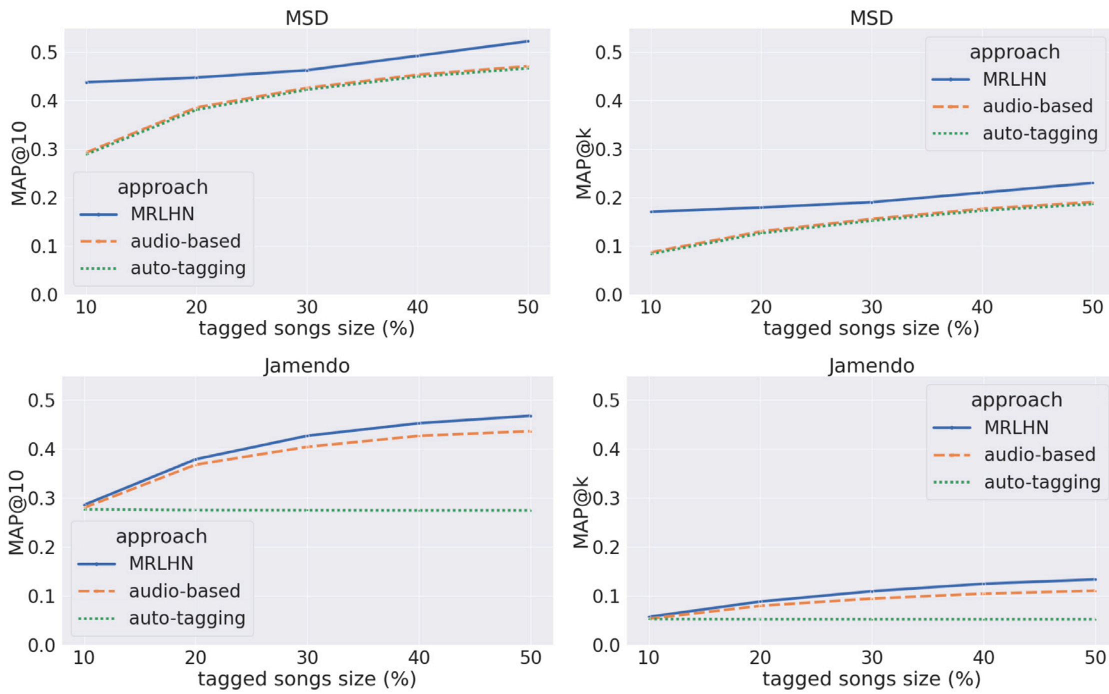

Fig. 2. The behavior of mean average precision of the approaches with the increasing percentage of untagged songs.

The MAP@k is a metric commonly used to evaluate the recommender algorithm and measure the accuracy of its indications. We can treat our music information retrieval problem as a ranking task, where the order of the songs in the rank has relevance. When we calculate the average precision (AP_k), we compute the precision of our algorithm, considering only one query at a time. By adopting the MAP@k, we reproduce an evaluation that computes the generalization power of the proposed algorithm by analyzing all the possible correct songs for a query.

We evaluated each method at five different levels of missing tags (percentage of songs without associated tags) for statistical analysis: 90%, 80%, 70%, 60%, and 50%. To compose the dataset at each level, we select the songs at random. We repeated this process ten times with different random seeds. Considering 200 tags and all the different levels, we evaluated each approach in a total of 10,000 queries.

4.2. Results and discussion

The experimental results are discussed considering three different aspects. First, we present and discuss a general comparison of the proposed MRLHN method in relation to the Audio-Based Similarity and Auto-Tagging approaches. Second, we discussed the impact of the α and β parameters used in constructing the heterogeneous network. Finally, the third aspect involves a discussion of the MRLNH's ability to deal with out-of-vocabulary tags.

The proposed auto-tagging is displayed as "MRLHN", while the approach using only the melspectrogram to represent the musical audio is shown as "audio-based", and "auto-tagging" refers to an approach to the representation formed by the Musicnn tags. The overall average precision of the models, ignoring the variation of the parameters, is presented in Table 1. When considering the standard deviation of precision, all models are within the same distribution. Still, the proposed approach has the highest mean in all metrics and the most significant difference in the MSD. Our approach presents results closer to audio-based, which matches the problem since the melspectrogram is used as a representation to measure the similarity between the songs. When looking at the auto-tagging results, we can assume that just adopting the tags' information as a feature, even if they are based on a transfer learning process, may not be enough to build a musical representation with good discriminative power. Exploring the multimodality of data, as discussed in Section 2, tends to enrich representations.

Table 1
Overall average precision and standard deviation organized by approach and ignoring parameter variations.

Approach	MSD		Jamendo	
	MAP@10	MAP@k	MAP@10	MAP@k
MRLHN	0.4726 ± 0.1555	0.1963 ± 0.1260	0.4027 ± 0.1899	0.1027 ± 0.0776
audio-based	0.4080 ± 0.1520	0.1493 ± 0.1059	0.3833 ± 0.1777	0.0885 ± 0.0717
auto-tagging	0.4039 ± 0.1499	0.1456 ± 0.1057	0.2751 ± 0.1859	0.0522 ± 0.0384

Next, we explore the variation of parameters that modify the percentage of tagged and untagged data, the size of the search space, and its impact on model accuracy. Fig. 2 shows the behavior of precision measures given the variation in the volume of tagged and untagged data in each dataset and the mean and standard deviation in Table 2. With more labeled data, it is expected that our proposal has connections between data, producing more direct links between tagged and untagged data that have more significant similarity, and therefore between untagged and untagged data, positively reflecting on the propagation of tags. For audio-based and auto-tagging, it is expected that be indexed with more tagged data, more samples to assist in the search, and better performance in information retrieval. Our approach has the best performance in both datasets, although, for Jamendo, we need more labeled data to overcome audio-based. On MSD, as it is a problem with less data volume, audio-based and auto-tagging represent the songs similarly when increasing the tagged data.

To analyze the method's performance with the variation in the number of retrieved songs, we must observe the results in the two metrics differently. For the MAP@10 metric, we will analyze whether expected music is retrieved in top-10 songs when a query is made. With an increase in the music list size, there is a probability increase in retrieving the expected musics. Another metric is the MAP@k, which computes precision considering that the retrieved musics are in the correct position according to the increase in size of the list. Thus, the increase of music retrieved tends to reduce the precision of the method.

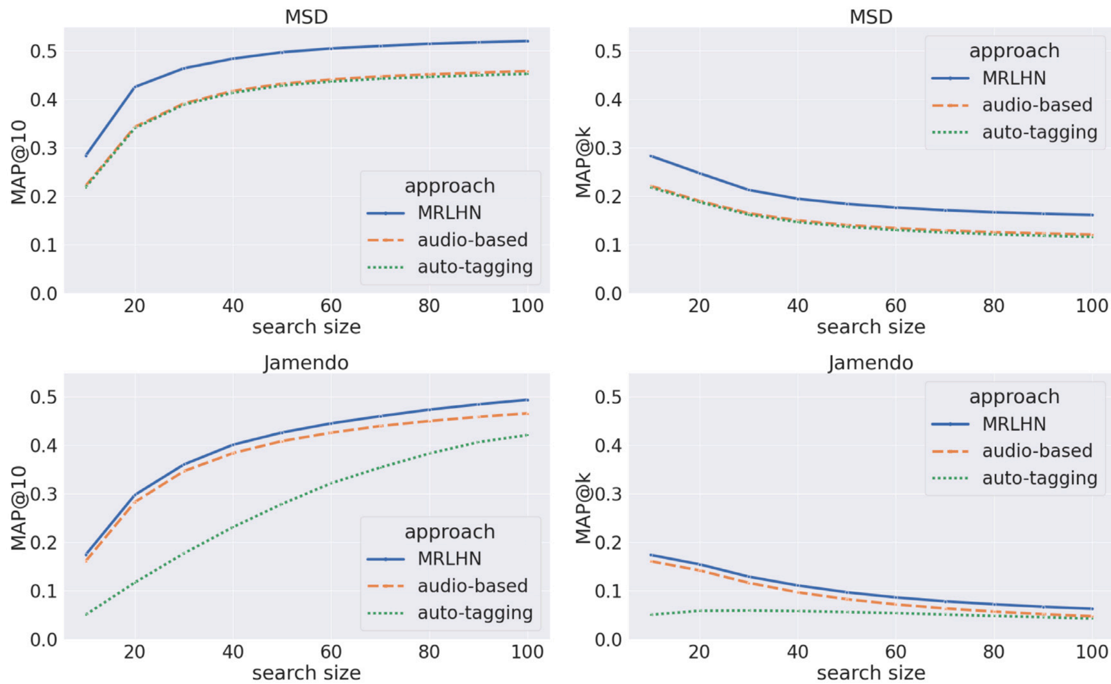

Fig. 3. The behavior of mean average precision of the approaches with the increasing song list size retrieved.

Table 2
Mean average precision for our approach MRLHN compared to the other approaches, considering the number of untagged songs.

Approach	MSD		Jamendo	
	MAP@10	MAP@k	MAP@10	MAP@k
MRLHN	0.4720 ± 0.0349	0.1964 ± 0.0237	0.4025 ± 0.0735	0.1027 ± 0.0308
audio-based	0.4012 ± 0.0758	0.1486 ± 0.0399	0.3832 ± 0.0634	0.0885 ± 0.0229
auto-tagging	0.3972 ± 0.0758	0.1449 ± 0.0397	0.2752 ± 0.0008	0.0522 ± 0.0002

Table 3
Mean average precision for our approach MRLHN compared to the other approaches, considering the song list size retrieved.

Approach	MSD		Jamendo	
	MAP@10	MAP@k	MAP@10	MAP@k
MRLHN	0.4720 ± 0.0729	0.1964 ± 0.0401	0.4016 ± 0.1005	0.1031 ± 0.0384
audio-based	0.4012 ± 0.0766	0.1486 ± 0.0298	0.3823 ± 0.0963	0.0889 ± 0.0394
auto-tagging	0.3972 ± 0.0759	0.1449 ± 0.0304	0.2741 ± 0.1273	0.0522 ± 0.0058

Table 4
Mean average precision for the impact of the α and β parameters used in constructing the heterogeneous network, to represents the amount of tagged and untagged songs and yours relationships.

	MSD					
	MAP@10			MAP@k		
α \ β	3	5	7	3	5	7
3	0.474520	0.475400	0.477985	0.197483	0.195939	0.198432
5	0.468189	0.473902	0.474786	0.194452	0.196713	0.197259
7	0.465454	0.470601	0.472206	0.193383	0.195580	0.196576
	Jamendo					
	MAP@10			MAP@k		
α \ β	3	5	7	3	5	7
3	0.400251	0.406208	0.407861	0.102804	0.104523	0.104787
5	0.399565	0.403522	0.403939	0.101583	0.102796	0.103122
7	0.396861	0.400913	0.403332	0.100222	0.101670	0.102331

The Fig. 3 shows the behavior of precision measures considering the variation in the size of the music retrieved in each dataset. In contrast, the mean and standard deviation is shown in Table 3. We can see again that our method performs better in all evaluated scenarios, and for both metrics, we have the expected behavior results. In both problems, a trend of stabilization in precision is considered as more information is retrieved. We also have an approximation trend for the Jamendo dataset, which allows us to consider the representations used to offer a clean comparison.

As we described in Section 3, the topology of our heterogeneous network is built from the relationship that is measured by the similarity of song audio content of tagged and untagged songs. In addition to presenting the behavior of approaches considering the overall percentage of tagged data, we also evaluate our method by diversifying the number of tagged nodes of a given tag, α, and the untagged nodes, β, which will be reached. In this scenario, we want to assess the MRLHN's ability to propagate tag embeddings to similar songs and the need for a larger volume of tagged data to build a consistent network. We show in Table 4 that the MRLHN's precision kept similar across all parameter variations. This stable behavior allows us to infer that the propagation of tag embeddings can reach the nodes even without many tagged songs. We also can highlight the capability of our approach to solving cold-start and long-tail problems.

No correlation was observed among the analyzed approaches between the frequency of a tag used as a query and the methods' performance. However, we analyze the quartiles of intervals ordered by tag popularity, calculated by the number of music the tags appear on. Although our method also presents the best performance in all scenarios, it is noticed that there is no defined pattern among a set of tags, see Fig. 4, which shows the ability to exploit problems with both characteristic of the long tail, as well as cold start, as it performs consistently in both scenarios.

Finally, in Fig. 5 we analyze a scenario where there are tags out-of-vocabulary, considering only the default Musicnn tags, which allows us to explore the limitations of pre-trained models that have fixed vocabularies. We highlight the performance of the auto-tagging method since the tags used here are the tags used for indexing this method. This scenario allows us to justify using BERT for textual representation

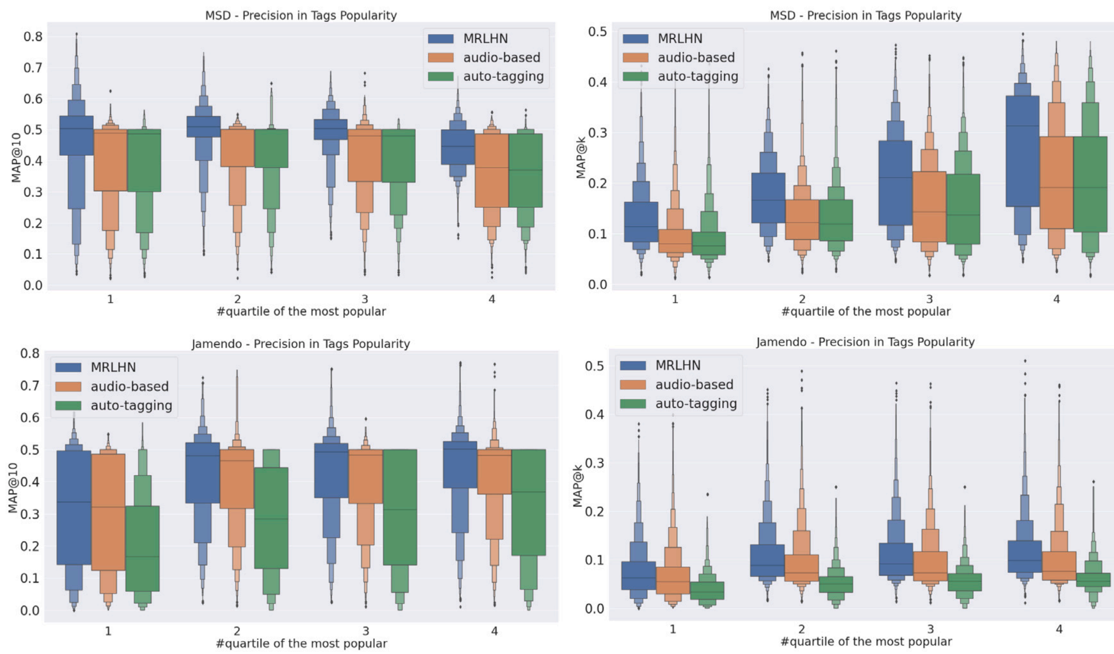

Fig. 4. Mean average precision distribution according to the popularity of the tags, organized by quartiles that concentrate the appearance frequency of the tags.

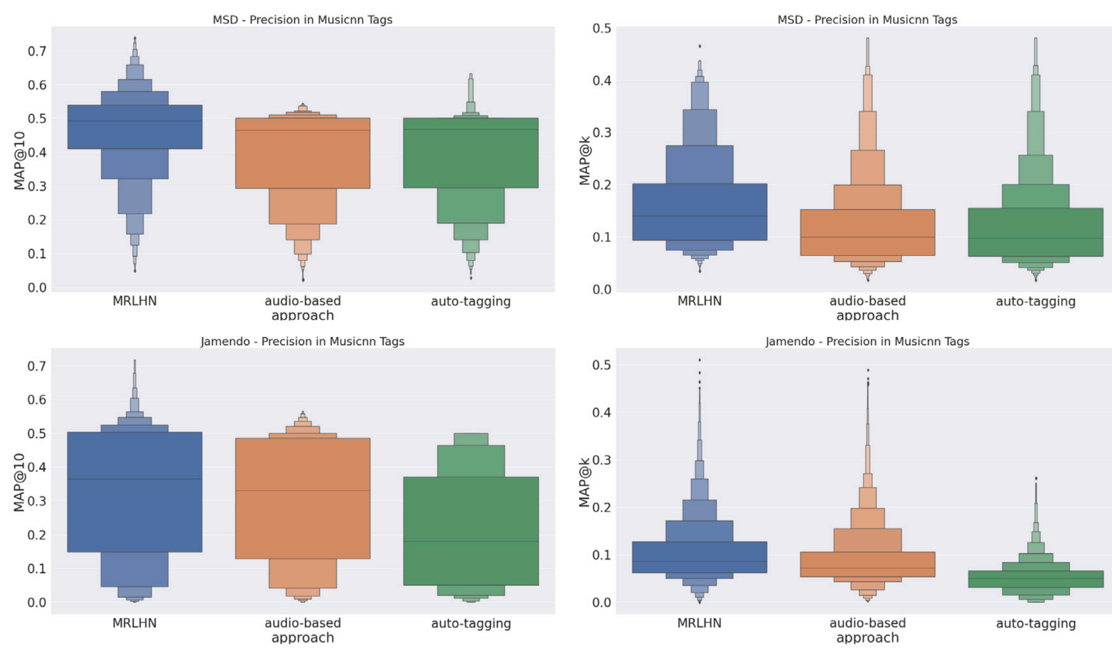

Fig. 5. Mean average precision distribution according to Musicnn tags, to represents the fixed vocabulary scenario.

so that even for an unknown tag, we can have a similar representation when considering the musical context. Using only the acoustic information, audio-based had a slightly lower performance than auto-tagging in MSD. However, in Jamendo and more extensive problems, audio-based stands out and allows us to infer the need for more features to build the data representation.

5. Conclusion

This work presents a new method that explores musical multimodal structure by incorporating tag and audio information simultaneously to learn a new representation. The proposed method is based on a heterogeneous network and solves the tag-based music retrieval problem, including long-tail or cold-start scenarios. Our approach consists of an initial representation formed by embeddings of the textual tags extracted by a state-of-the-art language neural model refined considering the constraints of the heterogeneous network topology constructed by the audio representation. We evaluated the method in a music retrieval task in different scenarios to analyze the precision according to the variation of the number of tagged songs, the size of the retrieved songs list, popularity of tags, and out-of-vocabulary tags. We can mitigate the absence and out-of-vocabulary of query tags with learned representation. We highlight the relevance of exploring multimodalities to complement a representation vector semantically that can improve the discrimination of music data independently of the problem scenario.

For future work, we will incorporate more features into the heterogeneous network construction process and refinement of the initial embeddings, such as lyric embeddings and other audio features. We will also assess the representation learned in multitask problems such as genre classification and rank prediction. Finally, to better assess the effectiveness of the proposed method, comparisons between deep neural models of feature fusion will be conducted.

CRediT authorship contribution statement

Angelo Cesar Mendes da Silva: Data curation, Conceptualization, Software, Writing – original draft. **Diego Furtado Silva:** Supervision, Writing – review & editing. **Ricardo Marcondes Marcacini:** Supervision, Writing – review & editing.

Declaration of competing interest

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